Coffee Quality Dataset - EDA Documentation

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1. Objective

The objective of this exploratory data analysis (EDA) is to uncover patterns, trends, and insights in the coffee quality dataset. The dataset contains information about various coffee samples, including sensory attributes (like aroma and flavor), bean characteristics, and farm metadata (such as country of origin and altitude).

2. Dataset Overview

The dataset includes the following types of data:

Quality Measures:

- Aroma
- Flavor
- Aftertaste
- Acidity
- Body
- Balance
- Uniformity
- Cup Cleanliness
- Sweetness
- Moisture
- Defects
- Cupper Points

Bean Metadata:

- Species (Arabica / Robusta)
- Processing Method
- Color

Farm Metadata:

- Country of Origin

- Region
- Altitude
- Owner
- Farm Name
- Mill
- Company

3. Data Cleaning

- Removed duplicate entries
- Checked for missing values and nulls
- Ensured categorical variables had consistent formatting
- Binned altitude values into ranges for grouped analysis

4. Univariate Analysis

Analyzed distribution of individual features:

- Quality metrics: Histograms showed most scores were skewed toward higher values
- Categorical variables: Count plots showed Arabica was more common than Robusta; certain countries dominated the sample

5. Bivariate Analysis

Explored relationships between two variables:

- Correlation matrix showed strong relationships between aroma, flavor, balance, and cupper points
- Boxplots revealed Arabica typically has higher quality scores than Robusta
- Scatterplots showed a positive correlation between altitude and certain sensory attributes (e.g., aroma)

6. Visual Insights

Included several informative plots:

- Bar plot: Average Cupper Points by Country of Origin
- Heatmap: Processing Method vs Species vs Cupper Points
- Stacked bar: Uniformity, Sweetness, and Clean Cups by Country

7. Summary of Findings

- Arabica coffee generally scores higher in quality metrics
- Countries at higher altitudes tend to produce coffee with better aroma and flavor
- Processing methods and species play a key role in determining overall cup quality

8. Conclusion

This EDA provided a comprehensive view of the coffee dataset, revealing key patterns and enabling data-driven recommendations for improving coffee quality sourcing and evaluation strategies.